

To maximize conversion and product yield while minimizing impurity formation across all three reactions

reactions

High Conversion
Fraction $([B] / [A]_0)$

High Selectivity
 $([B] / [I])$

Low impurity
Fraction $([I] / [A]_0)$

Objectives as Metrics

Time for equivalent yield can be an important economic factor, but not in our case due to emphasis on yield and impurity only in PS.

Approach

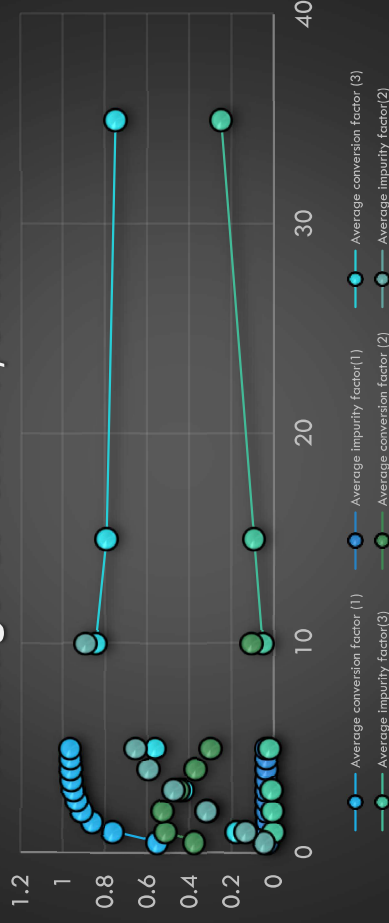
Fix stratrum: Fixing time and plotting average of time vs HF and IF at different concentration and temperature of all reactions.

Calculate and extract both the median and mean values of matrices for analysis.

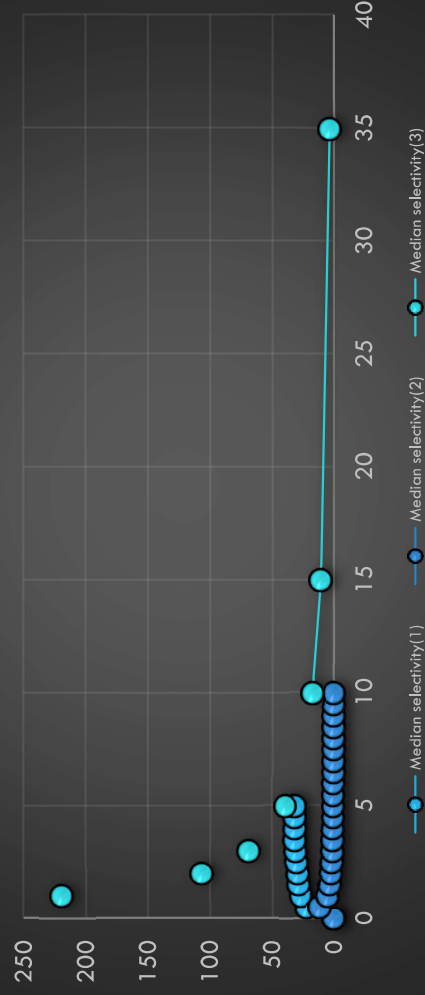
Plot graph, view trends and fit equations.

Rank equations on general trends of Metrics and verify with general points

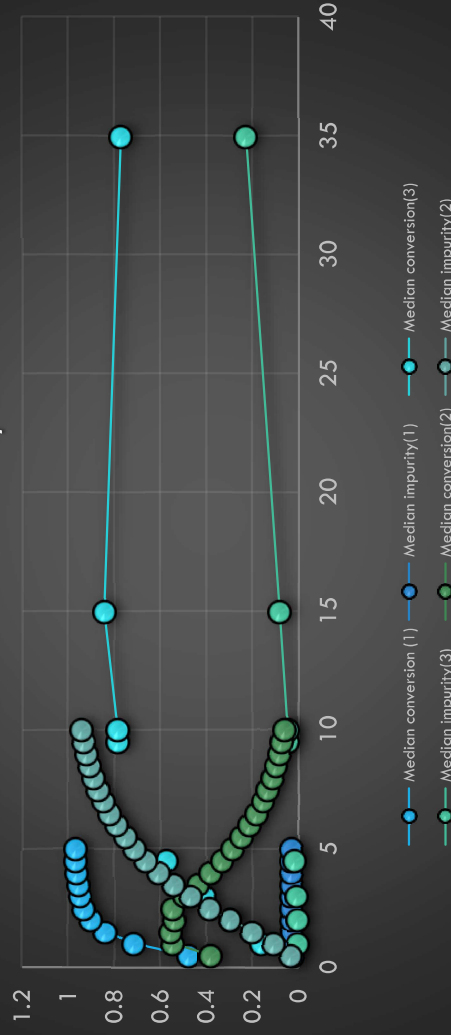
Average CF & IF V/S Time

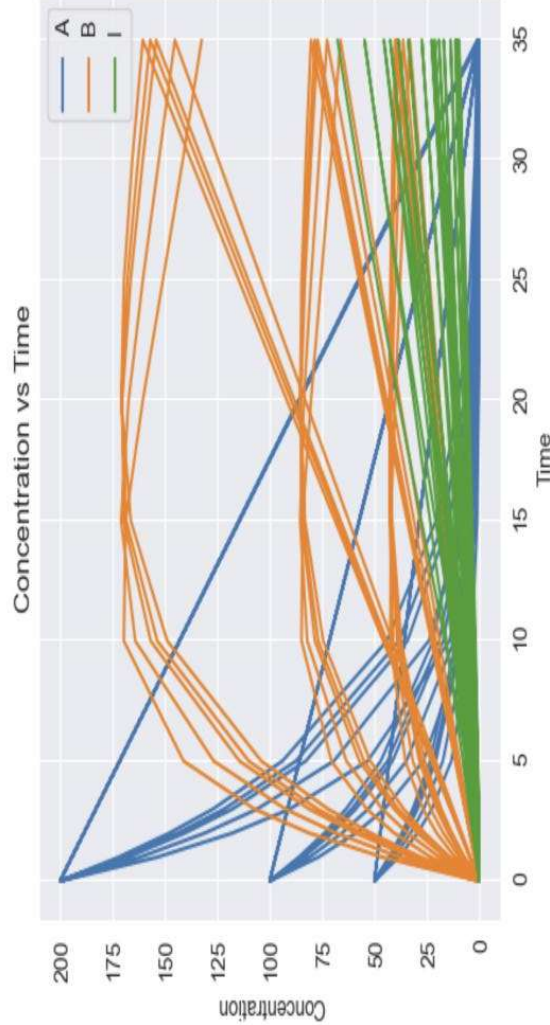
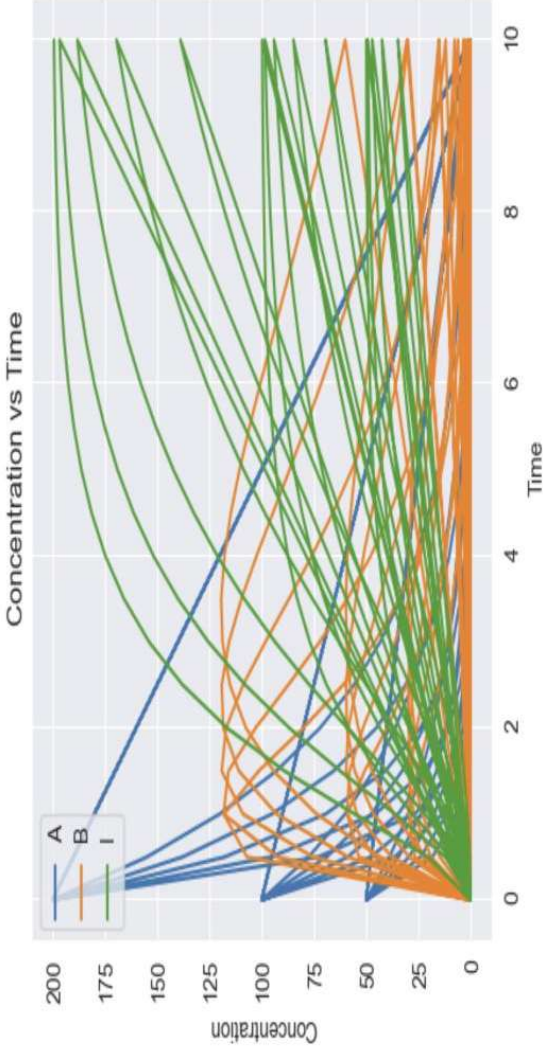
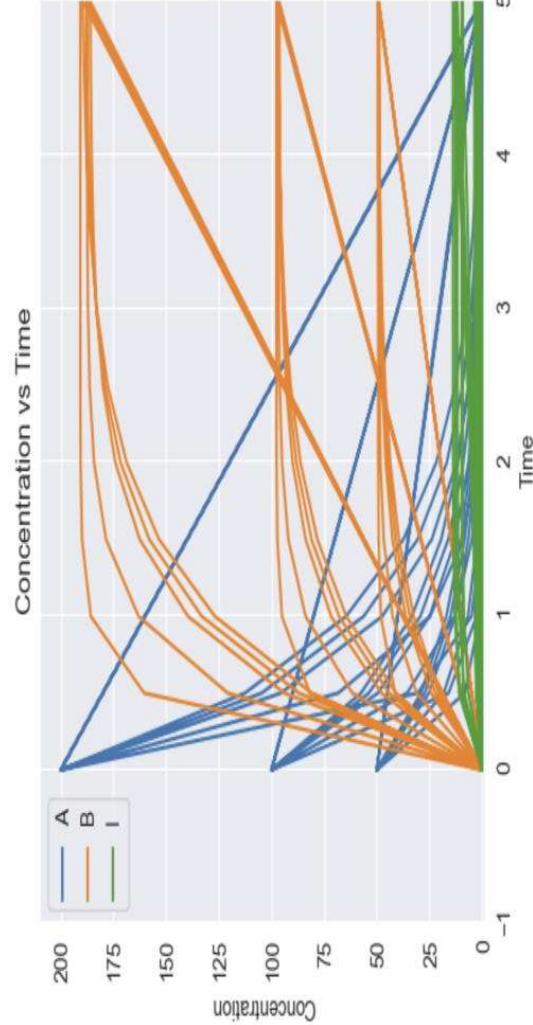


Median Selectivity V/S Time



Median Metrics V/S Time





Analysis of these Graphs

- Across all three reactions, trajectories **cluster** by **3 initial concentration** (50, 100, 200 mg/mL).
- These graphs look like that of **exponential**, due to immediate rise, decline and plateaus in graph.
- **Reaction-1:** A falls to near zero by ~3–5 h, B rapidly plateaus high, and I stays low and nearly flat making it the most suitable reaction
- **Reaction-2:** B shows a clear **intermediate-time peak** and then declines as I rises steadily to dominate by ~10 h making it worst choice of all.
- **Reaction-3:** Conversion is **slow**; B increases gradually up to ~0.75–0.95 over ~15–35 h while I grows modestly proving this to be a moderate suitable reaction.

Idea generation

While reviewing the data sets, we observed **acceptable variation** within each reaction at **fixed time** across **different temperatures** and **concentrations**, so we chose to use averaged values in analysis and concentration and impurity factor to further reduce the comparison stress as goal is only **to rank**.

Analysis based on average

Observations

- ✓ **Reaction 1** conversion climbs to 0.90-0.97 in (3-5 h) and remains flat, impurity stays 0.02-0.03 and nearly constant low, median selectivity holds around 30 which makes it well suited for flow reaction.
- ✓ **Reaction 3** conversion rises to 0.75-0.85 at (10-15 h) then eases to ~0.77 at 35 h, impurity increases from near zero to ~0.23 by 35 h, median selectivity declines from about 220 at 1 h to ~3 at 35 h which make it workable but slower and more sensitive.
- ✓ **Reaction 2** conversion decreases from ~0.54 at 1 h to about 0.06 at 10 h, impurity rises from about 0.11 at 1 h to about 0.94 at 10 h, median selectivity drops from about 11 at 0.5 h to about 0.06 at 10 h which makes it suitability least under these conditions.

Benefits of this analysis

Outlier Resilience

Mean and median balance extreme value influence.

Noise Reduction

Averaging minimizes noise for clearer data comparisons.



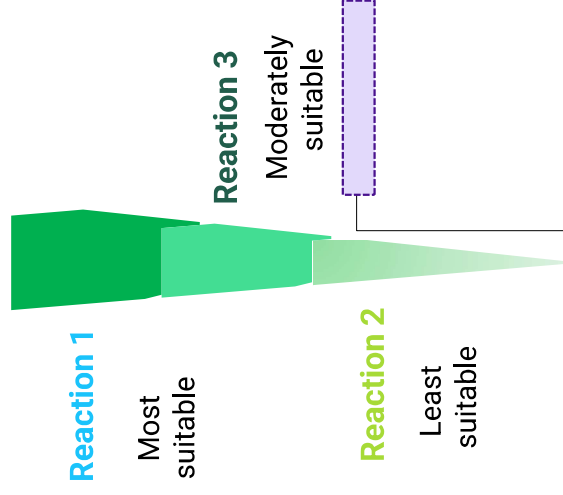
Single-Axis Evaluation

Focuses on time as the sole varying parameter.

Calculations

Conclusion

Suitability of Reactions for Flow Chemistry



Potential Problems

Averages Mask Complex Interactions and Biases.

Misleading Averages
Averages obscure underlying complexities.

Hidden Interaction
Averages smooth over true variable interactions($[A_0], T$)

Uneven data coverage
Biases average calculations.

Outliers Influence
Extremes disproportionately skew the average value.

Is there other approach ?

- ❖ We can try to analyse graph of overall data of reactions and pick a trend.
- ❖ We can try to fit equation in every dataset of all reactions

Time

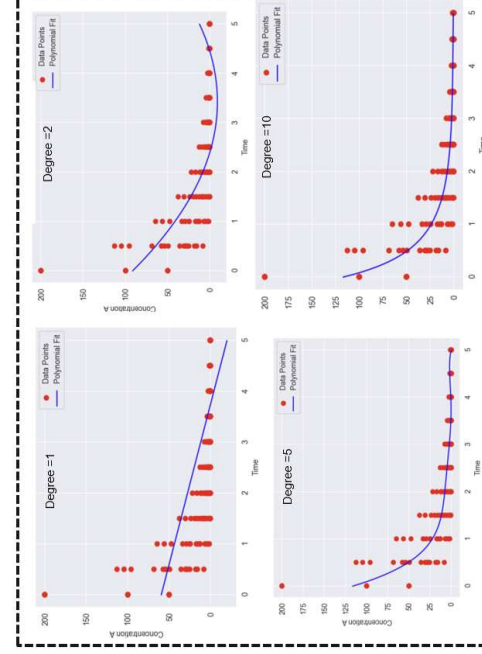
Did the first order come in our dream ?



We have **only one reactant A**, **without catalyst** mentioned, making it the most probable for first order.



- ❑ When we plot all points the **clusters** seem to **follow exponential** equation.
- ❑ When we try to fit **polynomial** function in model, it tends to **better** with each **increment in degree** with a lot outliers.



Observations for reaction 1,2 & 3

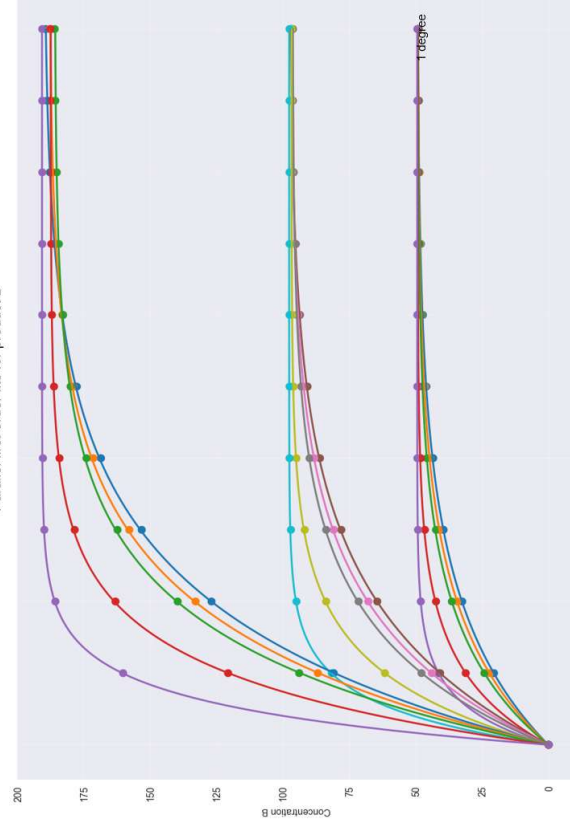
Reaction 1 seems to follow



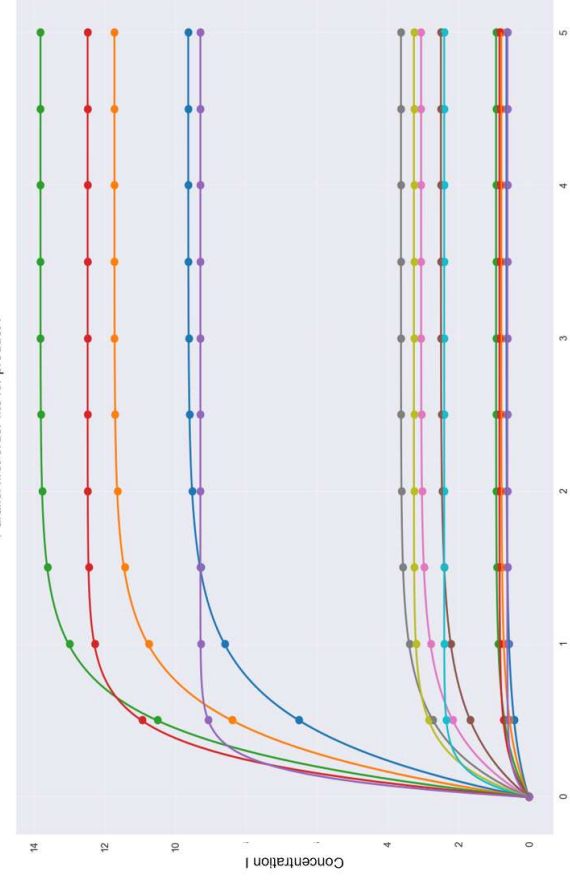
Reaction 2&3 seems to follow



Parallel first-order fits for product B

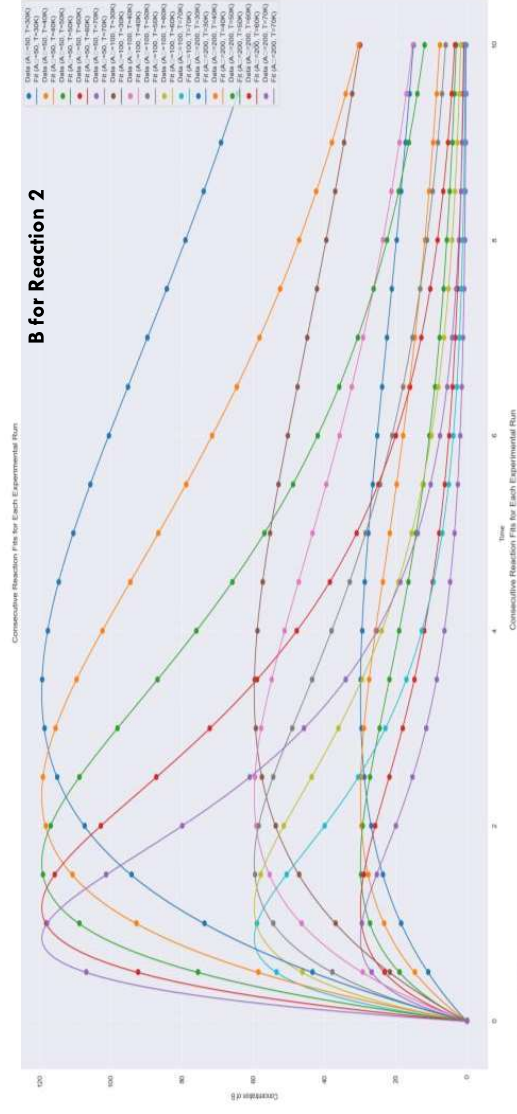


Parallel first-order fits for product I

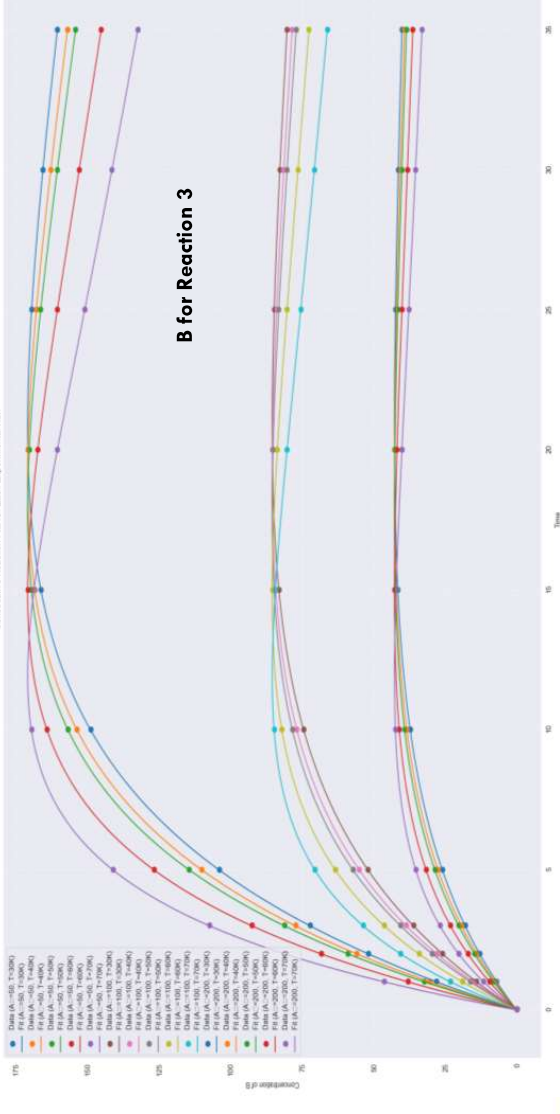
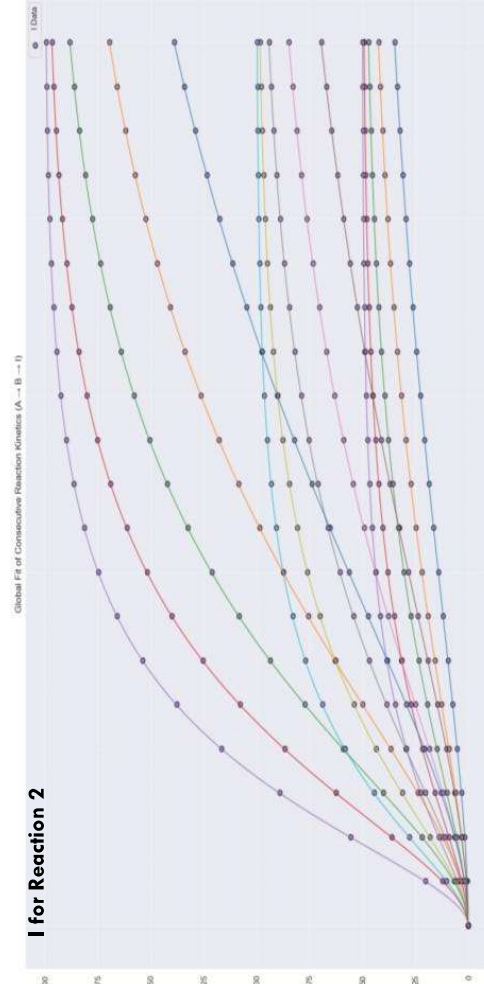


●	Data AD=50, T=30 K
●	Fit, k1=0.027, k2=2.130
●	Data AD=50, T=40 K
●	Fit, k1=0.037, k2=2.369
●	Data AD=50, T=50 K
●	Fit, k1=0.050, k2=2.660
●	Data AD=50, T=60 K
●	Fit, k1=0.067, k2=3.058
●	Data AD=50, T=70 K
●	Fit, k1=0.088, k2=3.584
●	Data AD=100, T=30 K
●	Fit, k1=0.054, k2=2.131
●	Data AD=100, T=40 K
●	Fit, k1=0.074, k2=2.365
●	Data AD=100, T=50 K
●	Fit, k1=0.100, k2=2.661
●	Data AD=100, T=60 K
●	Fit, k1=0.133, k2=3.046
●	Data AD=100, T=70 K
●	Fit, k1=0.175, k2=3.534
●	Data AD=200, T=30 K
●	Fit, k1=0.088, k2=2.126
●	Data AD=200, T=40 K
●	Fit, k1=0.147, k2=2.363
●	Data AD=200, T=50 K
●	Fit, k1=0.197, k2=2.654
●	Data AD=200, T=60 K
●	Fit, k1=0.261, k2=3.023
●	Data AD=200, T=70 K
●	Fit, k1=0.343, k2=3.571

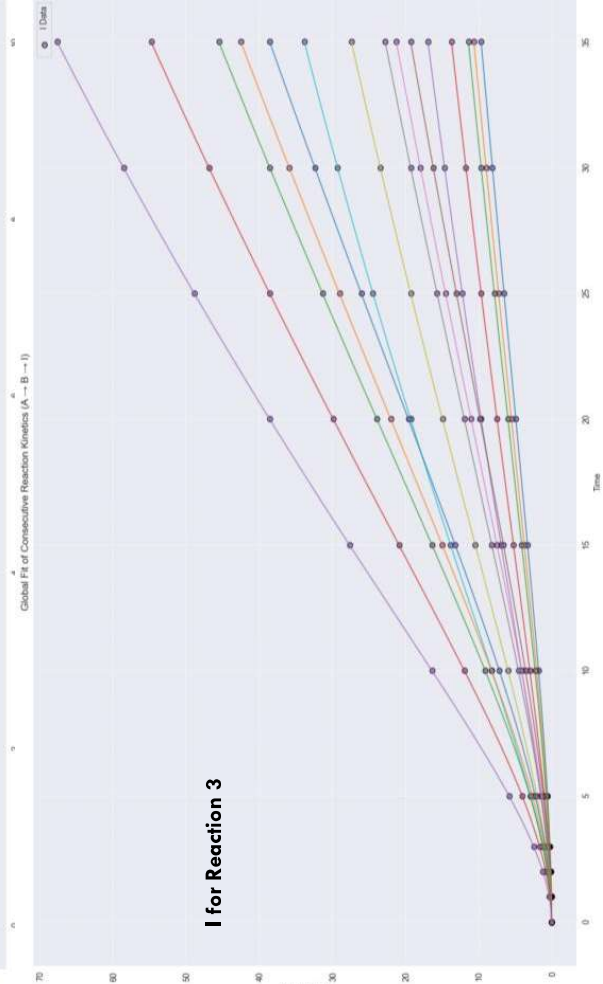
●	Data AD=50, T=30 K
●	Fit, k1=1.060, k2=0.014
●	Data AD=50, T=40 K
●	Fit, k1=1.177, k2=0.019
●	Data AD=50, T=50 K
●	Fit, k1=1.324, k2=0.026
●	Data AD=50, T=60 K
●	Fit, k1=1.967, k2=0.034
●	Data AD=50, T=70 K
●	Fit, k1=3.240, k2=0.044
●	Data AD=100, T=30 K
●	Fit, k1=1.055, k2=0.028
●	Data AD=100, T=40 K
●	Fit, k1=1.170, k2=0.038
●	Data AD=100, T=50 K
●	Fit, k1=1.314, k2=0.050
●	Data AD=100, T=60 K
●	Fit, k1=1.953, k2=0.066
●	Data AD=100, T=70 K
●	Fit, k1=3.230, k2=0.077
●	Data AD=200, T=30 K
●	Fit, k1=1.048, k2=0.055
●	Data AD=200, T=40 K
●	Fit, k1=1.157, k2=0.074
●	Data AD=200, T=50 K
●	Fit, k1=1.295, k2=0.098
●	Data AD=200, T=60 K
●	Fit, k1=1.926, k2=0.130
●	Data AD=200, T=70 K
●	Fit, k1=3.305, k2=0.171



I for Reaction 2



I for Reaction 3



Reaction 1

A0= 50,	T=30 K :	k1=1.0603	s ⁻¹
A0= 50,	T=40 K :	k1=1.1773	s ⁻¹
A0= 50,	T=50 K :	k1=1.3235	s ⁻¹
A0= 50,	T=60 K :	k1=1.9674	s ⁻¹
A0= 50,	T=70 K :	k1=3.5725	s ⁻¹
A0= 100,	T=30 K :	k1=1.0553	s ⁻¹
A0= 100,	T=40 K :	k1=1.1703	s ⁻¹
A0= 100,	T=50 K :	k1=1.3139	s ⁻¹
A0= 100,	T=60 K :	k1=1.9532	s ⁻¹
A0= 100,	T=70 K :	k1=3.5495	s ⁻¹
A0= 200,	T=30 K :	k1=1.0457	s ⁻¹
A0= 200,	T=40 K :	k1=1.1568	s ⁻¹
A0= 200,	T=50 K :	k1=1.2954	s ⁻¹
A0= 200,	T=60 K :	k1=1.9260	s ⁻¹
A0= 200,	T=70 K :	k1=3.5046	s ⁻¹

Reaction 2

A0= 50,	Temp = 30K =>	k1=0.5108,	k2=0.1532
A0= 50,	Temp = 40K =>	k1=0.7443,	k2=0.2233
A0= 50,	Temp = 50K =>	k1=1.0597,	k2=0.3179
A0= 50,	Temp = 60K =>	k1=1.4770,	k2=0.4431
A0= 50,	Temp = 70K =>	k1=2.0191,	k2=0.6057
A0= 100,	Temp = 30K =>	k1=0.5108,	k2=0.1532
A0= 100,	Temp = 40K =>	k1=0.7443,	k2=0.2233
A0= 100,	Temp = 50K =>	k1=1.0597,	k2=0.3179
A0= 100,	Temp = 60K =>	k1=1.4770,	k2=0.4431
A0= 100,	Temp = 70K =>	k1=2.0192,	k2=0.6057
A0= 200,	Temp = 30K =>	k1=0.5108,	k2=0.1532
A0= 200,	Temp = 40K =>	k1=0.7443,	k2=0.2233
A0= 200,	Temp = 50K =>	k1=1.0597,	k2=0.3179
A0= 200,	Temp = 60K =>	k1=1.4770,	k2=0.4431
A0= 200,	Temp = 70K =>	k1=2.0192,	k2=0.6057

A0=50,	T=30K =>	k1=0.5108,	k2=0.1532
A0=50,	T=40K =>	k1=0.7443,	k2=0.2233
A0=50,	T=50K =>	k1=1.0597,	k2=0.3179
A0=50,	T=60K =>	k1=1.4770,	k2=0.4431
A0=50,	T=70K =>	k1=2.0192,	k2=0.6057
A0=100,	T=30K =>	k1=0.5108,	k2=0.1532
A0=100,	T=40K =>	k1=0.7443,	k2=0.2233
A0=100,	T=50K =>	k1=1.0597,	k2=0.3179
A0=100,	T=60K =>	k1=1.4770,	k2=0.4431
A0=100,	T=70K =>	k1=2.0192,	k2=0.6058
A0=200,	T=30K =>	k1=0.5108,	k2=0.1532
A0=200,	T=40K =>	k1=0.7443,	k2=0.2233
A0=200,	T=50K =>	k1=1.0597,	k2=0.3179
A0=200,	T=60K =>	k1=1.4770,	k2=0.4431
A0=200,	T=70K =>	k1=2.0192,	k2=0.6057

A0= 50,	T=30 K :	k1=0.0273	s ⁻¹
A0= 50,	T=40 K :	k1=0.0374	s ⁻¹
A0= 50,	T=50 K :	k1=0.0502	s ⁻¹
A0= 50,	T=60 K :	k1=0.0669	s ⁻¹
A0= 50,	T=70 K :	k1=0.0884	s ⁻¹
A0= 100,	T=30 K :	k1=0.0544	s ⁻¹
A0= 100,	T=40 K :	k1=0.0744	s ⁻¹
A0= 100,	T=50 K :	k1=0.0999	s ⁻¹
A0= 100,	T=60 K :	k1=0.1327	s ⁻¹
A0= 100,	T=70 K :	k1=0.1751	s ⁻¹
A0= 200,	T=30 K :	k1=0.1077	s ⁻¹
A0= 200,	T=40 K :	k1=0.1471	s ⁻¹
A0= 200,	T=50 K :	k1=0.1970	s ⁻¹
A0= 200,	T=60 K :	k1=0.2611	s ⁻¹
A0= 200,	T=70 K :	k1=0.3446	s ⁻¹

Reaction 3

A0= 50,	Temp = 30K =>	k1=0.1514,	k2=0.0076
A0= 50,	Temp = 40K =>	k1=0.1656,	k2=0.0083
A0= 50,	Temp = 50K =>	k1=0.1766,	k2=0.0088
A0= 50,	Temp = 60K =>	k1=0.2119,	k2=0.0106
A0= 50,	Temp = 70K =>	k1=0.2649,	k2=0.0132
A0= 100,	Temp = 30K =>	k1=0.1514,	k2=0.0076
A0= 100,	Temp = 40K =>	k1=0.1656,	k2=0.0083
A0= 100,	Temp = 50K =>	k1=0.1766,	k2=0.0088
A0= 100,	Temp = 60K =>	k1=0.2119,	k2=0.0106
A0= 100,	Temp = 70K =>	k1=0.2649,	k2=0.0132
A0= 200,	Temp = 30K =>	k1=0.1514,	k2=0.0076
A0= 200,	Temp = 40K =>	k1=0.1656,	k2=0.0083
A0= 200,	Temp = 50K =>	k1=0.1766,	k2=0.0088
A0= 200,	Temp = 60K =>	k1=0.2119,	k2=0.0106
A0= 200,	Temp = 70K =>	k1=0.2649,	k2=0.0132

A0=50,	T=30K =>	k1=0.1514,	k2=0.0076
A0=50,	T=40K =>	k1=0.1656,	k2=0.0083
A0=50,	T=50K =>	k1=0.1766,	k2=0.0088
A0=50,	T=60K =>	k1=0.2119,	k2=0.0106
A0=50,	T=70K =>	k1=0.2649,	k2=0.0132
A0=100,	T=30K =>	k1=0.1514,	k2=0.0076
A0=100,	T=40K =>	k1=0.1656,	k2=0.0083
A0=100,	T=50K =>	k1=0.1766,	k2=0.0088
A0=100,	T=60K =>	k1=0.2119,	k2=0.0106
A0=100,	T=70K =>	k1=0.2649,	k2=0.0132
A0=200,	T=30K =>	k1=0.1514,	k2=0.0076
A0=200,	T=40K =>	k1=0.1656,	k2=0.0083
A0=200,	T=50K =>	k1=0.1766,	k2=0.0088
A0=200,	T=60K =>	k1=0.2119,	k2=0.0106
A0=200,	T=70K =>	k1=0.2649,	k2=0.0132

Classification of reactions

How can we even think of that ?

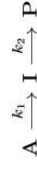
Reaction 1

- **Impurity remains very low compared to product**, with B/I ratios consistently high across time.
- **Selectivity stays stable**, showing only minor drift even at higher concentrations.
- **Impurity does not increase with time**. It plateaus early, while product keeps rising, indicating no significant $B \rightarrow I$ pathway.
- **Kinetic signature matches parallel first-order with $k_1 \gg k_2$** . It shows conversion is mainly $A \rightarrow B$, with only a small side-channel to I.

Reaction 2 & 3

- **Impurity** rises **steadily** and significantly with time, showing onward conversion of **B**.
- **Selectivity declines** at longer times, unlike reaction-1 where somewhat it stays stable.
- **B increases** initially as **A is consumed**, then reaches a **peak**. After the peak, **B decreases** steadily because it is further converted to impurity I.
Justifying $\mathbf{A \rightarrow B \rightarrow I}$
- **Clusters of graph** show different relation from parallel first order reaction.

Overall Analysis



$$\frac{d[A]}{dt} = -k_1[A]$$

$$\frac{d[\text{I}]}{dt} = k_1 [\text{A}] - k_2 [\text{I}]$$

$$\frac{d[\text{P}]}{dt} = k_2 [\text{I}]$$

$$[A] = [A]_0 e^{-k_1 t}$$

$$[\mathbf{I}] = \frac{k_1[\mathbf{A}]_0}{k_2 - k_1} (e^{-k_1 t} - e^{-k_2 t})$$

$$[\text{P}] = [\text{A}]_0 \left[1 + \frac{1}{k_1 - k_2} (k_2 e^{-k_1 t} - k_1 e^{-k_2 t}) \right]$$

Formulation



$$\frac{d[A]}{dt} = -k_1[A] - k_2[A] = -(k_1 + k_2)[A]$$

$$[\mathbf{A}] = [\mathbf{A}]_0 e^{-(k_1 + k_2)t}$$

$$\frac{d[B]}{dt} = k_1 [A] = k_1 [A]_0 e^{-(k_1 + k_2)t}$$

$$[\text{B}] = -\frac{k_1[\text{A}]_0}{k_1 + k_2} \left(e^{-(k_1 + k_2)t} \right) + c_1$$

$$c_1 = \frac{k_1[A]_0}{k_1 + k_2}$$

$$[\text{B}] = \frac{k_1[\text{A}]_0}{k_1 + k_2} (1 - e^{-(k_1 + k_2)t})$$

$$[C] = \frac{k_2[A]_0}{k_1 + k_2} (1 - e^{-(k_1 + k_2)t})$$

Reference

Analysis and Results

- **Reaction 1** : $k_1 \gg k_2$ ($\sim 1.0\text{--}3.5 \text{ s}^{-1}$ vs $0.01\text{--}0.17 \text{ s}^{-1}$) across T, B rises to near plateau; **I plateaus early** stays low (no $B \rightarrow I$ decay). High productivity makes it the **best** choice for our reaction.
- **Reaction 2** : $k_1 \sim 0.5\text{--}2.0$ vs $k_2 \sim 0.15\text{--}0.60 \rightarrow k_1 \approx k_2$ across. In a consecutive system, **$A \rightarrow B \rightarrow I$** system, **B is quickly drained to I & selectivity collapses** as time increases. Makes it the **worst** choice
- **Reaction 3** : Selectivity potential is good ($k_1 \gg k_2$) initially, but decrease with time. The reaction is **slow** overall, so **long time** is required for a yield which **fade after the peak** and **I still rises at late times, cleaner than R-2** and

Approach of Selection

❖ We will use Fractional-Factorial i.e corners for main effects.

- Covers the design space via four extreme (corner) points in dataset
- Yields orthogonal main effects and estimate data with minimal variance
- Maximizes information per run for fast and high-confidence screening
- Outliers are explained and included

Selected Data Points



70 °C, 200 mg/mL, 1 h

High temperature
High concentration
Short time

70 °C, 50 mg/mL, 5 h

Low temperature
High concentration
Long time

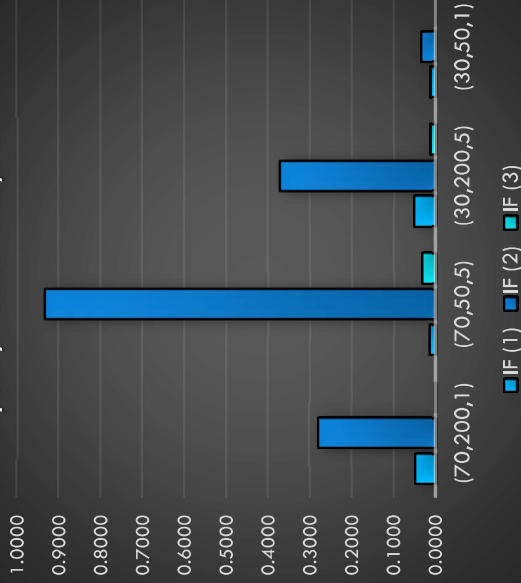
30 °C, 200 mg/mL, 5 h

High temperature
High concentration
Short time

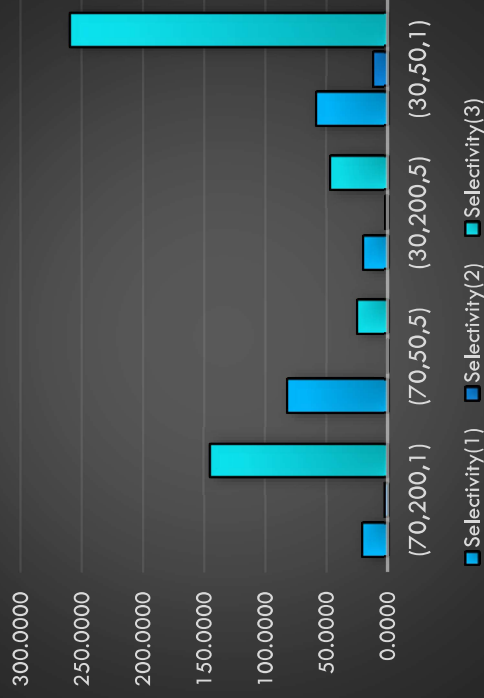
30 °C, 50 mg/mL, 1 h

Low temperature
Low concentration
Short time

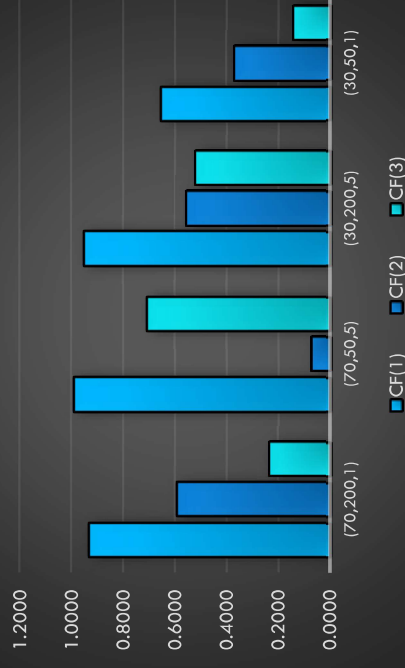
Impurity Factor Analysis



Selectivity Analysis



Conversion Factor Analysis



Observations

- ❑ **Reaction-1** shows **consistently high conversion** ($\approx 0.95+$), **very low impurity** (≤ 0.1), and maintains **good selectivity**.
- ❑ **Reaction-2** is highly impurity-prone, with impurity surging at High-T/Long-t (70,50,5) and High-C/Long-t (30,200,5), while its conversion rises higher than Reaction-3 at Low-T/Short-t (30,50,1) but with **very poor selectivity**.
- ❑ **Reaction-3** shows **moderate conversion** but **very low impurity**, delivering the **best selectivity** (peak at 30,50,1).

Verification

Across the four corners, **R-1** shows the highest conversions ($\approx 0.95+$) with consistently low impurity (≤ 0.1); **R-3** has moderate conversion but the cleanest profiles (highest selectivity); **R-2** suffers large impurity spikes at (70,50,5) and (30,200,5) and only a **brief conversion edge** at (30,50,1) with **poor selectivity**. Therefore, the data support the rank: **R-1 > R-3 > R-2**.

Reactor

Set up : A jacketed tubular plug-flow reactor fed by a fast micromixer

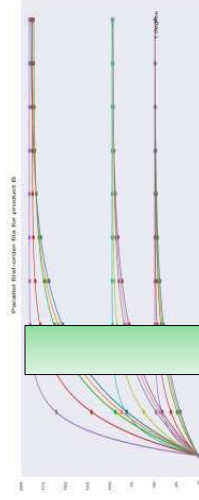
Impacts : Plug Flow Reactor narrows **Residence Time Distribution (RTD)** and limits over-holding of product, which generally **decreases impurity** from consecutive/parallel paths and **improves selectivity** at a given conversion where impurity somewhat increase with time(our case).

Faster micro mixing **reduces local hotspots and side reactions in the reactor**, giving **lower early time impurity and more reproducible conversion.** [Ref 1](#)

Time and control

Set up : Targeting the suitable time at flat CF plateau

Impacts : Shorter time **reduces impurity** from $A \rightarrow I$ and $B \rightarrow I$ but may **lower conversion**. Longer time **raises conversion** but **increases impurity** when side paths exist. Choose time to sit in our favorable window like ~2.5 - 3 hour in our case.



Dealing with small conc.

when we carefully look into the graph and dataset, we can see impurity increase for increase in concentration. So dealing with small concentration can be helpful in order to reduce impurity

Hydrology

Set up : Installation of **static mixers**(single phase homogeneous)

Impacts : Static mixers split/recombine laminar layers, eliminating micro-gradients that seed impurity pathways. They are a standard, passive way to intensify mixing in laminar flow. [Ref 1](#) [Ref 2](#)

It gives **better selectivity** at a given conversion. Static mixing reduces axial dispersion and helps tubular reactors approach plug-flow behavior, which is known to favor selectivity. [Ref 1](#) [Ref 2](#)

Enhanced radial mixing by static mixers increases heat removal effectiveness, limiting hot spots that can grow I. [Ref 1](#)

Temperature and pressure

Set up : Fit a **Back-Pressure Regulator (BPR)** and run **isothermally**

Impacts : Installing BPR **prevents boiling** and allows **safe superheating** of the solvent for rate control without two phase formation in our homogeneous mixture. [Ref 1](#) [Ref 2](#)

Counter pressure from the BPR **suppresses** bubble formation maintaining stable RTD & selectivity. [Ref 1](#)

Precise isothermal control hotspots, reduces thermal gradients, improving selectivity and handling of exotherms. [Ref 1](#)

Stable isothermal operation gives consistent performance and smoother scale-up to manufacturing. Also, our graph seem to vary a lot with changes in temperature.

Inline monitoring

Set up : Place an **ATR-IR (or UV-vis)** flow cell after the hot zone

Impacts : Track **A, B, and I** in real time and maintain the setpoint (e.g., CF ~0.95, IF ~0.03) by small adjustments to T or flow. [Ref 1](#)

Inline IR/UV/NMR are well-established for continuous monitoring and control in flow. [Ref 2](#)

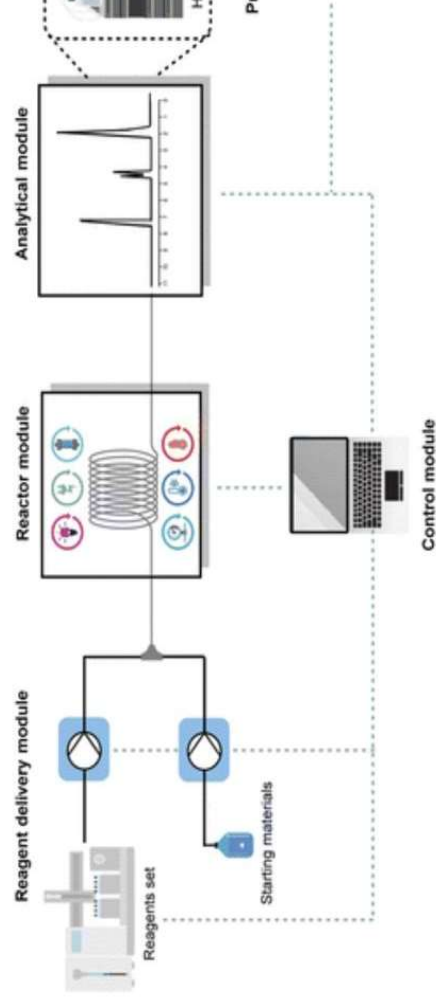
Continuous monitoring can help to navigate and control most of the aspects in this reaction.



FlowIR device



UV-Vis device

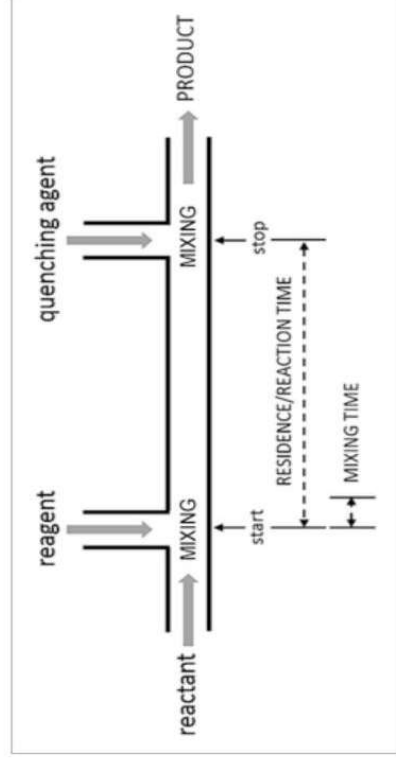
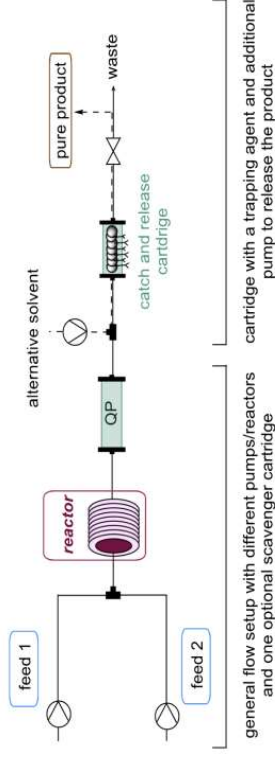


Hard stop and work-up

Set up : **Immediate quench** in a chilled tee

Impacts : The quench freezes the B:I ratio at the optimum residence time, while **LLE** and **scavengers** remove species (residual A, catalysts, acids/bases) that could **promote downstream I formation** during hold-up or work-up. [Ref 2](#)

Scavenger cartridge to remove residual A or catalyst that could form I downstream. [Ref 1](#)

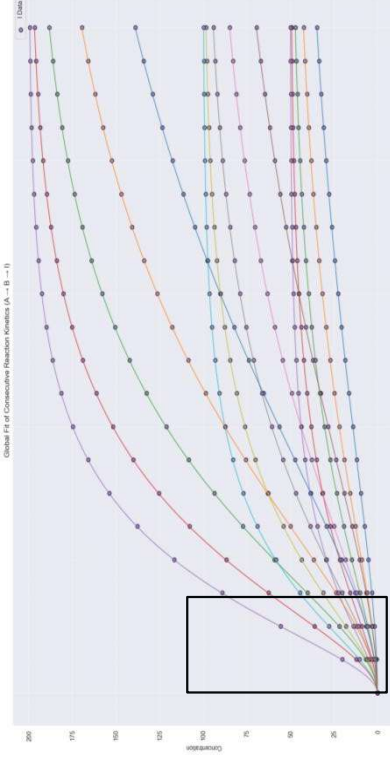


To propose modifications in reaction 2 to make it more amenable to flow processing

Shorten residence time(τ) and tighten the RTD

Move to a **micro-PFR** with **fast micro mixing** and intentionally **short τ** ($\leq 1-2$ h to match our early low-IF region).

Use similarly static mixers to give **better selectivity after micro mixing** at a given conversion



Distributed addition

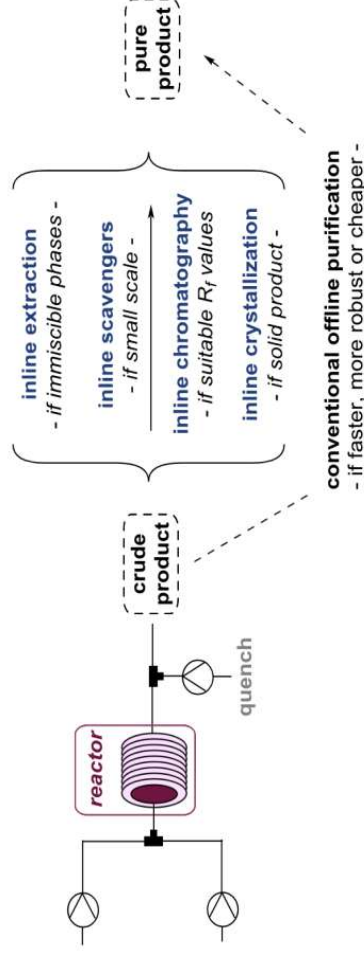
- Instead of charging all reagent at the reactor inlet, **split the limiting reagent** into 2-4 smaller feeds injected at intervals along the **PFR coil**; use **tee junctions** or side ports. This keeps the **instantaneous [A] or [B] lower**, smoothing out hot-spots. [Ref 1](#)
- Downstream inlets and telescoping** are standard in continuous synthesis and are effective for controlling competing pathways.
- By avoiding high local concentration, the **rate of side-reactions ($A \rightarrow I$ or $B \rightarrow I$)** is suppressed. The main pathway ($A \rightarrow B$) still progresses, but competing channels get starved of excess substrate.

Lower temperature to penalize impurity channels

We can see in our graphs, the impurity is generally increasing on increment of temperature which means the side path has a **higher apparent E_a (activation energy)**. **Lowering T** can improve the B:I rate ratio even if overall rate drops. If rate becomes limiting, compensate safely in flow by **superheating** under a BPR (moderate pressure) or **numbering-up** channels rather than raising T too far. [Ref 1](#)

Immediate quench & continuous removal of drivers

Put a **cold quench tee** at the **reactor exit** and couple it to **membrane L-L separation** to remove promoters that convert $B \rightarrow I$. This reduces any post-reactor growth of I in hold-up lines and lets us chain a second, milder polishing step if needed. [Ref 1](#)



Calculations

<https://1drv.ms/x/c/7ba247a4ada7bc9a/EdIZDM39ueROqFMhSol3HQEBhfvynttXlo5VRU6EXYQWbw?e=IBVQwu>

<https://1drv.ms/x/c/7ba247a4ada7bc9a/EcoXXmMi5pdEkyb1oinb s8BfepULNIHqTUBPh3OrnjV9Q?e=xJqRxi>

<https://1drv.ms/x/c/7ba247a4ada7bc9a/EW8sxNCyPoNBreA4d01K0hEBlynmrCHH8jDyBKk2nnFx7A?e=KOMh6z>

<https://drive.google.com/drive/folders/1GFIHTNM7b Ug w9nsXDdCEAZnkpD0t8l?usp=sharing>